

Teleosemantics and the Critique of Adaptationism

It HAS LONG BEEN ASSUMED that there is some (presumably strong) connection between teleological theories of representation and adaptationism in evolutionary biology, but just what that connection is has never been clearly explained. Thus, it is often assumed that any attacks on adaptationism also constitute attacks on teleological theories of representation (or *teleosemantics*, for short). Determining exactly what the connection between these two things is will be the burden of this paper. Before even beginning to try to see what the relationship between adaptationism and teleosemantics is, it will be

helpful to get a rough idea of what the *relata* are. So, roughly, *teleosemantics* is a teleofunctional account of what determines the content of linguistic and mental representations. Thus, teleosemantics determines the content of a representation by determining the representation's function—what it is “supposed to” represent, where the “supposed to” is a *biological* “supposed to”. An object's teleofunction is what it has been *selected* to do, what it has been “designed for”. So, the teleofunction of an object is a function that does not rest on current properties, but instead depends upon that object's history, in particular, its selective history. The most sophisticated version of teleosemantics is found in MILLIKAN 1984, see also MILLIKAN 1993, PAPINEAU 1987, and DENNETT 1995. *Adaptationism*, unlike teleosemantics, has been defined and characterized primarily by those who

Abstract

The point of this essay is to clarify the relationship between adaptationism (and criticisms thereof) and teleosemantics. In order to do this, two questions are discussed: (1) How correct as a methodology does adaptationism have to be to support teleological semantics; and (2) Is adaptationism that correct? In answering the first question, I argue that a teleological account of public language can be separated from a teleological account of mental representations. The teleosemantic account of public language that I suggest is entirely independent of adaptationism, while that of mental representations is importantly dependent upon it. In answering the second question, I examine a few of the more well-known criticisms of adaptationism (problems with optimization, “free rider” criticisms) to determine whether they are successful enough to make a teleosemantics for mental representations impossible.

Key words

Meaning, teleosemantics, MILLIKAN, adaptationism.

wish to criticize it. Adaptationism, sometimes also called “functionalism” (see GOULD 1991), is, again roughly, the tendency of theorists in evolutionary biology to consider organisms as sets of traits, each of which has been optimally designed by natural selection to serve its function. This tendency has been famously ridiculed by GOULD/LEWONTIN 1979 as the “Panglossian paradigm”, see also GOULD/VRBA 1982, KITCHER 1985, and for a defense of adaptationism DENNETT 1983, 1995.

The connection between teleosemantics and adaptationism can be brought somewhat into

focus now. Here it is in a first, rough pass: “Adaptationism is the tendency to assume that all of the individual parts of an organism were selected for, and so have *biological functions*. If, as some have claimed recently, adaptationism is bad methodology, then many parts of living organisms might not have functions. The parts of humans that underly their linguistic and cognitive skills may be among those that do not have functions. So teleosemantics, which determines the content of linguistic and mental representations according to their functions, could be a misguided enterprise.”

Notice that this attempt at spelling out the connection is littered with ‘might’s and ‘could’s. In what follows, I will try to replace these modal words with ones that are more straightforwardly assertoric. To do so, of course, will require considering teleose-

mantics, adaptationism and a few prominent critiques of it in some detail. Here, then, is the plan. First we will consider the occasionally hairy details of teleosemantics, and then, more briefly, what adaptationism amounts to, with the goal of answering the question:

(1) How correct does adaptationism need to be for teleosemantics to be possible?

Next, we will examine some of the more prominent lines of criticism of adaptationism and attempt to determine whether adaptationism is as true as is deemed necessary by our answer to question (1). Question (2), then, is:

(2) And is adaptationism *that* correct?

Before beginning to try to answer these two questions, however, it is worth pointing out that *philosophical* considerations that might cause problems for teleosemantics, like those raised in FODOR 1990, will not be considered here. There are two reasons for this: first, if it turns out that teleosemantics does not survive the *empirical* challenges posed by the foes of adaptationism, any philosophical considerations in its favor are moot; second, FODOR's criticisms have been fairly decisively rebutted by the proponents of teleosemantics (see MILLIKAN 1993 and DENNETT 1995) and so are not really worth considering here.

We shall begin our consideration of question (1) by examining MILLIKAN's teleosemantic theory. Put very roughly, according to MILLIKAN, a thing A has a *proper function*¹ F only if it meets one of the two following conditions:

(1) A originated as a reproduction of some prior item or items that, due in part to possession of properties reproduced, have actually performed F in the past, and A exists because of this performance, or (2) A originated as the product of some prior device that, given its circumstances, had performance of F as a proper function and that, under those circumstances, normally causes F to be performed by means of producing an item like A. (1993, pp13–14).

So, basically, an object has a proper function only if it has been *copied* because of something it can do (satisfying the first clause) or is produced by something with a proper function (satisfying the second clause). MILLIKAN's term *reproductively established family* covers both these cases (1984, pp23–25) So from now on, I will refer to things that were copied or produced by something copied as *members of reproductively established families*. By claiming that to have a function an object must be a member of a reproductively established family, MILLIKAN makes her definition of function different than those of WRIGHT 1973, CUMMINS 1975 and BIGELOW/PARGET-

TER 1987. This is the case because the fact that to have a function something must be a member of such a family makes it the case that an object's functions cannot be determined by merely examining that object's *current* dispositional properties. Instead, the item's function depends upon how and from what it was reproduced, that is, its *history*. In particular, an item depends upon its *selective history*, for otherwise, items which are copied but do not intuitively have functions will count as members of reproductively established families. As MILLIKAN puts it, "Only if an item or trait has been *selected* for reproduction, *as over against other traits*, because it sometimes has a certain effect does that effect count as a function." (1993, pp35–36) So, for example, full (or half or new) moons do not have a function, even though they are members of a reproductively established family, because they were not *selected for*—the moon does not become full (or half or new) because it is somehow better for the moon to be full (or half or new) than to be in another phase. Finally, then, an object has a proper function if and only if it is a member of a reproductively established family and has been selected for. MILLIKAN claims that this definition covers bodily organs, instinctive behaviors, learned behaviors, reasoned behaviors, customs, language devices, as well as conscious intentions and purposes. (1993, p14)²

Now if the critiques of adaptationism to be considered were able to show that nothing has a function in MILLIKAN's sense, we would need delve no further into teleosemantics. They do not, however, and do not even purport to do so³. We will, therefore, press on into the details of MILLIKAN's account. The goal of teleosemantics is to determine the content of *representations* via analysis of their teleofunctions. Thus we must now consider MILLIKAN's theory of representations. Representations are a variety of what she calls *intentional icons*, which, she claims, are distinguished by the following four characteristics:

(1) An intentional icon is a member of a *reproductively established family* and was selected for, thus it has proper functions.

(2) In biologically normal cases (cases similar to those in which the icon has historically fulfilled its function), an intentional icon stands between two cooperating devices, one which produces the icon and one which consumes it. These two devices are *designed* or *standardized* to fit one another.

(3) In biologically normal cases, the intentional icon serves to adapt the consumer device to conditions such that functions of that device can be performed under those conditions.

(4) It is a proper function of the intentional icon to make changes in the consumer in accordance with certain mapping rules. It is the way that the environment must be in order to accord with these mapping rules that determines the *content* of the intentional icon. (1984, pp96–97)⁴

These four features of intentional icons are such that they can be fulfilled *separately* by both linguistic objects, such as sentences of a public language, and mental representations. In other words, it could be the case that sentences of a public language are representations that can be given a teleosemantic account, while mental representations cannot have a teleosemantic account, or vice versa. Furthermore, it could be that both can receive such an account, but *not the same account*.

This is a crucial feature of teleological theories of content. In all such theories, the meaning of sentences is not dependent upon any relation that sentences might have to mental representations. As MILLIKAN puts it, “Thus the intentionality of a public–language sentence is not derived from the *intentionality* of the inner representations that it [biologically n]ormally produces or expresses. Sentences are *basic* intentional items. And they are intentional for reasons that can be described without raising or answering any questions about what the mental is as such.” (1984, p90)

DENNETT, one of the other main proponents of teleosemantics, has made the same claim in many different venues, for example in his disagreements with John SEARLE’s contention that artifacts such as computers can only have *derived intentionality*. DENNETT claims that there is no good distinction between real meaning or intentionality and derived meaning or intentionality. See DENNETT 1995, ch.14 for an example of the denial of this distinction in connection to teleosemantics.

Because sentences and other language devices do not derive their content from mental representations, it is at least a *conceptual possibility* that sentences are intentional icons for reasons independent of the reasons that mental representations are intentional icons. This point is somewhat surprising and crucial for the purposes of this paper. It is *surprising* because it leads to the fact that the meaning of sentences we utter is not necessarily present to consciousness, nor can we know for certain that our sentences mean anything at all. The claim is *crucial* to the purposes of this paper for reasons that will be demonstrated below. So rather than just claim that it is the case that the meaning of language devices can be separated from the meaning

of mental representations, I will prove that it is so. To do so, I will now give an independent teleosemantic account of, first, sentences, and then, mental representations.

Consider a sentence S, “I’d like some pancakes.”

(1s) A token of S is reproduced (that is, repeated) by individual speakers because it causes certain responses in hearers. Namely, the hearers are led to believe that the speaker is in a pancake–desiring state and/or brings it about that the speaker gets some pancakes. Thus S is a member of a reproductively established family (via the first part of the definition). Furthermore, S was selected for, as against other sentences such as S’, “That house is on fire”, because of certain effects that S produces in hearers, which S’ does not. Thus S has a function. So, the first clause of the definition of intentional icons is satisfied.

(2s) The sentence, in normal situations, stands between a speaker and a hearer, who are standardized to fit one another—they speak the same language. This satisfies clause (2).

(3s) The sentence, in normal cases, serves to adapt the hearer so that the hearer can perform some of his or her functions. In the case of S, the hearer may be caused to believe that the speaker desires pancakes or, if the speaker is a waiter or waitress, he or she may bring the speaker some pancakes. This satisfies clause (3)⁵

(4s) The speaker learns to reproduce tokens of the sentence S in certain situations, those in which she or he wishes to produce beliefs or actions in the hearer that correspond to the content of S. That is, S has as its function the production of certain states or actions in the hearer, such that those states correspond to the speaker’s actually wanting some pancakes. Thus S’s function is to map certain environmental conditions, in particular, those that correspond to its content.

Because it satisfies these four clauses, S is an *intentional icon*. Notice, again, that the story told to show that S fulfills these conditions, and so has content, makes no reference to mental representations. A similar story can, of course, be told for any sentence S. Thus, it could be that sentences have teleosemantic content, while mental representations do not.

The story for mental representations is exactly parallel, but it occurs *entirely within the head*. Consider the mental representation M whose content is “There’s food over there”.

(1m) M is produced by a *representation producer*, whose function it is to produce mental representations. The representation producer is part of an or-

ganism's anatomy (presumably, part of its brain) that was reproduced by natural selection because of its ability to produce items like M when environmental circumstances call for it. Thus, M is a member of a reproductively established family (by the second clause of the definition). It was selected for, as against some other variety of representation producer that produces M' (whose content is, say, "The house is on fire.") when environmental circumstances call for M. Thus, M has proper functions and satisfies clause (1) of the definition of intentional icons.

(2m) After being produced, M stands between the representation producer and a *reproduction consumer* (in biologically normal cases). The producer and consumer have been designed by natural selection to cooperate, to "speak the same language".

(3m) In normal cases, M causes a change in the representation consumer such that the consumer can fulfill some of its functions with respect to the conditions indicated by M. For example, M might cause a change in the consumer so that it can initiate motion towards the food. Thus, clause (3) is fulfilled.

(4m) The organism is configured such that its representation producer normally produces M, rather than M', only when environmental circumstances accord with the content of M. This is the case because producing M in when it is true that "There's food over there" causes the representation consumer to enter a state so that it behaves appropriately with respect to the content of M. So the function of M is to cause the representation consumer to enter a state that corresponds to its content by some certain mapping rules. Thus clause (4) is fulfilled by M.

Thus, mental representations are intentional icons and can have teleosemantic content. To the extent that these two separate teleosemantic accounts (1s–4s and 1m–4m) are reasonable, both sentences and mental representations can have contents that are determined by their teleofunctions. Furthermore, they can have them independently. I have labored this point for a reason. The independence of the teleosemantics of language devices and mental representations makes it the case that our initial question (1) will require two separate answers. *That is, the question "How correct does adaptationism need to be for teleosemantics to be possible?" will get one answer for language, and another for mental representations.* Before giving these answers, it will be useful to see in a little more detail what adaptationism is.

Adaptationism, as I mentioned earlier, seems as if it would not be especially easy to characterize because it was initially described by GOULD/LEWONTIN

(in 1979) who wished to criticize it, and thus was from its initial definition something of a caricature of the practice of actual evolutionary biologists. GOULD/LEWONTIN (1979) claim that adaptationist theorizing generally proceeds in two steps. First, "[a]n organism is organized into 'traits' and these traits are explained as structures optimally designed by natural selection for their functions." (1979, p77) And second, "After the failure of part-by-part optimization, interaction is acknowledged via the dictum that an organism cannot optimize each part without imposing expenses on others. The notion of trade-off is introduced and organisms are interpreted as best compromises among competing demands. Thus interaction among parts is retained completely within the adaptationist program. An suboptimality of a part is explained as its contribution to the best possible design for the whole." (1979, p75)

Surprisingly, DENNETT, in his book length defense of adaptationism (1995), endorses a very similar definition of it, proposed later by LEWONTIN. "[Adaptationism] is defined by one of its most eminent critics as the 'growing tendency in evolutionary biology to reconstruct or predict evolutionary events by *assuming* that all characters are established in evolution by direct natural selection of the most adapted state, that is, the state that is the optimal 'solution' to a 'problem' posed by the environment." (LEWONTIN 1983) These critics claim that, although adaptationism plays *some* important role in biology, it is not really all that central or ubiquitous—and, indeed, we should try to balance it with other ways of thinking. I have been showing, however, that it plays a crucial role in the analysis of every biological event at every level from the first self-replicating macromolecule on up." (DENNETT 1995, p238)

So, given that both the most eminent critics and the most ardent defender of adaptationism accept the same definition of adaptationism, we will also accept it. Thus, adaptationism is a methodology⁶ whereby one treats an organism as a set of traits, each of which is *selected for* because it is the optimal structure available to selection for the fulfillment of its function (see also the essays collected in ROSE/LAUDER 1996).

Now, finally, we can see the connection between adaptationism and teleosemantics. If adaptationism is bad methodology, that is, if most features of current plants and animals were not selected for, then most features of current plants and animals *might not have functions* according to MILLIKAN's definition. I say that they *might not* have functions as opposed to saying

that they *do not* have functions because adaptationism is a methodology for understanding natural selection, but things could have been selected for by agents. An object might be a member of a reproductively established family and have been selected for, thereby having a function, in a way that has nothing to do with evolution by natural selection. The account of sentence meaning described in (1s)–(4s) above is just such an account. Evolution by natural selection is not mentioned or relied upon there at all. So here is the answer to the first part of question (1):

Adaptationism could be entirely incorrect—it could be false that anything has a function derived from natural selection—and teleosemantics would still be possible for sentences and other language devices.

This answer makes sense when we consider that sentences and other language devices are *tools*, artifacts designed by people to serve certain purposes. Just as it is unproblematic to assign functions to my computer, independently of the correctness of adaptationism, (1s)–(4s) above show that it is also unproblematic to assign (teleo)functions to sentences.

It is worth pointing out that MILLIKAN would disagree with this answer to the first question. As we can see from MILLIKAN's claim that the meaning of language is basic, not derived, she would certainly acknowledge the *possibility* that language devices get their meaning differently than mental representations do. But she believes that *in fact* sentences are able to perform their functions because they cause mental representations in the hearer. Thus she would claim that question (1), "How correct does adaptationism need to be for teleosemantics to be possible?", should only get one answer. And, as we shall now see, it is not the same one that I have given in the case of language devices.

Unlike the story told for sentences, the account of the teleosemantics of mental representations offered above (1m–4m) does make ineliminable reference to objects and mechanisms that evolved by natural selection. The representation producer and the representation consumer were both "designed" by natural selection to have the functions that they do: the producer was selected for its ability to produce representations that map features of the environment according to the rules that the consumer uses; the consumer was selected for its ability to use the representation produced by the producer to initiate appropriate behavior with respect to the content it (the representation) maps. So, for teleosemantics for mental representations to be possible, it must be the case that certain portions of the brain were selected

for their abilities to produce and consume representations in the same language. Thus, we have the answer to the second part of question (1): *For teleosemantics of mental representations to be possible, adaptationism must be correct enough that the portions of the brain that produce and consume representations were selected for their abilities to do so, and, so, have producing and consuming representations as their functions.* MILLIKAN realizes that this is true, and since she thinks that there needs to be a unified account of linguistic and mental representation, she claims that she "needs to assume the truth of evolutionary theory in order to show that quite mundane functional items such as screwdrivers and kidneys are indeed items with proper functions." (1993, p16) By separating the teleofunctional account of language from that of mental representations we have shown that this is not necessarily the case.

We now have our answer to both parts of question (1), and can proceed to attempt to answer the second question:

(2) Is adaptationism correct enough to support teleosemantics?

Part of the answer to this question comes immediately. Since the teleosemantic story told above for *sentences* makes no reference to functions derived by natural selection, the fate of adaptationism can have no effect whatsoever. *So, whatever our verdict on adaptationism as a methodology, it is trivially sufficient for the teleosemantics of linguistic representations.* The pressing question, then, is "Is adaptationism sufficiently correct for teleosemantics of mental representations?" For it to be so, it must be the case that the critiques of adaptationism that we will now consider do not make it unreasonable to think that certain parts of the brain have producing and consuming representations as their proper function. These critiques will come in two main types—problems with optimization and evolutionary free riders.

The first line of criticism of adaptationism concerns the assumption made by adaptationist evolutionary biologists that natural selection is guaranteed to *optimize*, that is, to produce a feature of an organism that is ideally suited to perform some function made necessary by the organism's environment. It turns out that this assumption is simply false. KITCHER (1985) points out several ways in which natural selection will fail to yield an optimal solution to the "problem" posed by the environment. First, he describes the course of evolution at a locus in a Bantu population. The combination of alleles available in this population and their relative fitnesses combine so as to lead to the demise of the

fittest combination of alleles. (See KITCHER 1985, pp215–216) Furthermore, there are several other ways in which the most fit combination of alleles just cannot be reached by the population, KITCHER calls this the *trajectory problem* (“Intuitively, the idea is that you can’t get there from here.” 1985, p216) Such trajectory problems can be caused by a variety of factors such as small populations, various vagaries of chance, and interactions among gene loci. When there are trajectory problems, the best “solution” that can be hoped for is a *local* optimum, which is best described by altering KITCHER’s description of the trajectory problem—intuitively, the local optimum is “the best you *can* get to from here”. And it is often the case that once the population settles into this local optimum, the global optimum is no longer an available option. Thus, the adaptationist is wrong in assuming that natural selection is guaranteed to optimize.

What concerns us, remember, given the failure of natural selection to guarantee optimality, is the effect that this might have on the likelihood that portions of the brain might have representation production and consumption as their functions. It seems that the failure of guaranteed optimization has no effect on this. These parts of the brain, supposing that they exist, will still have been reproduced because of their ability to produce and consume representations. And it is still likely that they were selected for their ability to do so, even if they are only locally optimal for so doing. In other words, the fact that there might be, somewhere in the vast space of possible brain configurations, some unavailable configuration that would be even better at producing and consuming representations, does nothing to make it the case that the existing structures do not have functions. Furthermore, even if the existing representation producers and consumers are not the local optimum, this does nothing to change the fact that it is overwhelmingly likely that they were selected for their abilities. Thus, this first line of criticism of adaptationism fails to cause any difficulties for teleosemantics.

We must now consider the other main line of criticism of adaptationism: free riders, cases where selection *of* objects, rather than selection *for*, certain traits has occurred. SOBER (1984) has characterized this distinction in an illuminating way. He asks us to imagine a toy with balls of different colors and sizes (which are correlated in that every ball of a certain size is the same color) inside a cylinder that is divided into several chambers. Each divider between two chambers has holes of a particular size, with the largest holes between the top and second chamber, the

next largest holes between the second third and so on all the way down (see SOBER, 1984, p99, for a picture of this toy). With such a toy, if all the balls are placed in the top chamber and the toy is shaken, the balls will eventually descend to the lowest chamber with holes above it that they can fit through. Suppose, as SOBER does, that only the green balls are small enough to fit through all the holes and reach the bottom chamber.

“When the green balls reach the bottom more frequently than the blue ones, we think that there must have been a reason why. Green balls were selected; so they must have had some property that was selected for. But the property in question was not their color. There was *selection of* green objects, but no *selection for* greenness. I offer the following slogan to summarize this logical point: ‘selection of’ does not imply ‘selection for’.” (SOBER 1984, p100)

What this example shows is that since it is only properties that can be *selected for* and any object has many properties, there are likely to be evolutionary *free riders*, properties of objects of which *selection of* occurred, but without *selection for* the particular property in question. The greenness of the balls that were selected is just such a case—there was selection of the green objects, but this selection was selection for size, not greenness. Evolutionary free riders figure importantly in two separate critiques of adaptationism associated with Stephen Jay GOULD (GOULD/VRBA 1982, GOULD/LEWONTIN 1979). We shall now turn our attention to these.

GOULD/VRBA (1982) introduces a new term to evolutionary biology. *Exaptations*, they say, are characteristics that “evolved for other purposes (or for no function at all), and [are] later ‘coopted’ for their current role.” (1982, p6) So exaptations begin as free riders, and only later become useful for some purpose or other. Adaptationism is incorrect, they say, because it fails to consider that certain traits might be exaptations. To return to SOBER’s toy example, imagine that the greenness of the balls, which, remember, came along as a free rider when the smallness was selected for, suddenly becomes quite a useful trait. Imagine, for example, that the balls are seeds and birds in their environment are, for whatever reason, particularly good at spotting green seeds, which they then ingest and excrete at some distant points, thereby spreading these green seeds, and the plants they grow into, widely. There was *selection for* the smallness of the seeds, and *selection of* the seeds themselves, including such traits as their greenness. Thus, GOULD/VRBA would say, the smallness of a seed is an adaptation, the greenness is an exaptation. Since ad-

aptations have functions and exaptations only have *effects*, the greenness of the seeds has no function⁸ GOULD/VRBA even suggest that the cognitive and language-using abilities of humans may be underwritten by exaptations in a brain whose features were selected for quite different reasons.⁹ This, then, is a direct assault on teleosemantics of mental representations. For if the features of brains that underly our representation producers and consumers are exaptations, if they exist in current humans as free riders that came along with some other trait that was selected for, then they can have no functions. So, teleosemantics for them is impossible.

The GOULD/VRBA critique, however, does not lead us to this conclusion. This is the case because all current adaptations are (or once were) exaptations—they only *begin* as free riders. A thought experiment from DENNETT 1995 can be used to make this point nicely. Imagine that some American engineers invent a device to recognize U.S. Treasury twenty-five cent pieces in American vending machines. Call such a device a “two-bitser” and imagine that it is (re)produced and placed in most vending machines. Each “two-bitser,” then, is a member of a reproductively established family and was selected *for* (by manufacturers of vending machines) because of its ability to recognize quarters, so its function is to do just that. Now imagine that after years of working in the United States, a vending machine that contains a two-bitser is brought to Panama, where there is a coin called (this is true, DENNETT tells us) a “quarter Balboa” that is the same shape and weight and approximate value as U.S. quarters. The two-bitser, it turns out, is also an excellent detector of quarter-balboas, and is used as a “q-balber” for several years. It is then an *exaptation* for the detection quarter balboas. So, GOULD/VRBA would say, it has as an *effect* the detection of quarter balboas, but its *function* is to detect U.S. quarters. (see DENNETT, 1995, 404–407) But now, going a small step beyond DENNETT’s thought experiment, imagine that a Panamanian manufacturer decides to build lots more of these devices precisely because of their ability to detect quarter balboas. The device now meets the conditions of having acting as a “q-balber” as its function; so, it is now also an *adaptation* for detecting quarter balboas.

What this shows is that adaptations must all have been features of an organism that were already present for some other (or no particular) reason, and were then pressed into ser-

vice in a certain job. In other words, all adaptations must have begun as exaptations. So either there are no exaptations (because once they are reproduced *because* they can perform the effect they were exapted for, they become adaptations, and so are no longer exaptations) or exaptations have functions (because adaptations do and all exaptations are also adaptations). Neither of these two options can lead to difficulties for the teleosemantics of mental representations. This is the case because parts of the brain that act as representation producers and consumers, if such there be, might be exaptations, but exaptations have functions too.

The final critique of adaptationism we will consider, and the first that might lead to some problems for the teleosemantics of mental representations, is another free rider criticism—GOULD/LEWONTIN (1979). In this masterpiece of scientific rhetoric¹⁰, GOULD/LEWONTIN decry the excesses of what they term the “adaptationist program”—a name which has stuck. They claim that adaptationism is a bad scientific methodology because its practitioners tell evolutionary “Just-so Stories”—assuming that a trait is the optimum solution to an environmental problem, and imagine scenarios whereby natural selection “found it”—to explain the presence of particular traits, ignoring the possibility that the traits could be free riders of one sort or another. GOULD/LEWONTIN do not primarily protest the fact that adaptationists *prefer* to tell these stories, rather than examine non-selective factors such as pleiotropy, linkage, genetic drift, allometry and developmental constraints (although they are concerned with this). Instead they are concerned about what they call “Panglossianism”, the assumption that *every* trait must be optimized for the environment, even in the face of disconfirmation of their Just-so Stories.

“We would not object so strenuously to the adaptationist program if its invocation, in any particular case, could lead in principle to its rejection for want of evidence. We might still view it as restrictive and object to its status as an argument of first choice. But if it could be dismissed after failing some explicit test, then alternatives would get their chance. Unfortunately a common procedure among evolutionists

does not allow such a definable rejection for two reasons. First, the rejection of one adaptive story usually leads to its replacement by another, rather than to a suspicion that a different kind of explanation might be required. ... Second,

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the criteria for acceptance of a story are so loose that many pass without proper confirmation. Often, evolutionists use *consistency* with natural selection as the sole criterion and consider their work done when they concoct a plausible story. But plausible stories can always be told." (1979, p79)

The point of the critique is that Panglossian adaptationists assume that there *must* be a selectionist history for every trait, even if the ones thought of thus far have been empirically disconfirmed and no others are in the offing. This critique of adaptationism is absolutely correct; evolutionary biologists, especially the sociobiologists who are the primary target of the argument, often do tend to ignore non-selectionist explanations of features of current plants and animals. This does not mean, however, that all or most or even many adaptationist explanations are incorrect. It also does not mean telling selectionist stories for traits is usually bad methodology or that natural selection is not the most important factor in evolution (see p81). All the GOULD/LEWONTIN critique shows is that we need to be *cautious* adaptationists, always remembering that there are alternatives to selectionist histories.

This, however, is enough to cause concern for the proponent of teleosemantics of mental representations. We can see this if we consider again what the theory requires. For there to be a teleosemantic theory of mental representations, it must be the case that (1) there are portions of the brain that produce and consume representations and (2) it must be their function to do so. (2) requires that there be an adaptive story, whereby the representation producers and consumers have been selected for their ability to trade in representations. And this is where the problems arise. The problems arise here because the teleosemantic story requires a selective history be told for a pair of brain structures *that could very well not exist*. That is, representation producers and representation consumers are theoretical entities; no one is certain that there are such things. Indeed if current explanations offered in connectionist artificial intelligence (RAMSEY 1997), dynamical systems theory (VAN GELDER 1995) and artificial life (WHEELER 1996) are generalizable to full explanations of cognition, cognition in general may be non-representational. So to invent an adaptationist story explaining why representation consumers and producers are they way they are is to take part in the Panglossian excesses that GOULD/LEWONTIN so rightly criticized. Thus, unless there is real confirmation that there are portions of the brain devoted to the production and consumption of representations (if there ever is such), we can-

not have a teleosemantics of mental representations because it would require that we tell a Just-so Story for two merely theoretical traits, which is *incautious* adaptationism, to say the least.

It is now time to take stock. What we have seen is that our first question—how correct does adaptationism need to be for teleosemantics to be possible?—requires a separate answer for public language and private mental representations. In the case of public language, we can have a teleosemantics without adaptationism being correct at all. It could even be the case that *no* part of humans evolved by natural selection, yet teleosemantics for public language would still be possible. For inner mental representations to get a teleosemantics, however, it must be the case that certain parts of the brain evolved by natural selection and so have functions. To claim that this is the case, even when there is no direct evidence that there are these parts, is Panglossian bad methodology of the kind that GOULD/LEWONTIN rightly object to. Thus we cannot conclude that representation consumers and producers have proper functions, which they must have for teleosemantics to be possible. Although adaptationism is correct enough for a teleofunctional account of sentences and the like, it fails to be so for mental representation.

The problem we have seen here is that teleosemantics of mental representations requires that the brain evolved in a particular way; it requires a Just-so Story for objects that *might not exist*. These objects might not exist because it is less than certain that what the brain does is manipulate representations. Although it is difficult to doubt that the brain is the organ that organizes and coordinates the activities that constitute thinking or intelligent behavior in some broad sense, it is certainly possible (even likely, perhaps) that most of what we call thinking is not the manipulation of representations. Perhaps our ability to do business in representations at all is dependent upon the fact that we all learn public languages. Thus we ought to be skeptical of the claim that the brain (or parts of it) is an adaptation for producing and consuming representations; we ought, also, to be skeptical of theories of meaning that depend upon this claim.

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Notes

- 1 In this paper, I will use 'function', 'proper function', and sometimes even 'teleofunction' interchangeably. Anything that I will claim has a function (simpliciter) satisfies MILLIKAN's definition of proper function, and proper functions are a variety of teleofunctions.
- 2 This however is not quite the case. This definition fails to cover newly invented devices that have not (yet) been reproduced. The same holds for neo-logisms which seem to have stipulated meaning even if never spoken again. This fault could be rectified by adding another clause to the definition.
- 3 This is the case because we will be considering what might be called revisionist critiques in this paper. More revolutionary critiques would argue that the problems with evolutionary biology are deeper than the rampant adaptationism that GOULD and LEWONTIN and VRBA decry. Examples of more revolutionary critiques include SAGAN/MARGULIS 1991, BUSS 1987, and some of the essays collected in ROSE/LAUDER 1996.
- 4 Representations are different from other intentional icons in that they are used in inferences. Henceforth, we will not worry about this difference. (see MILLIKAN 1993, pp106–116)
- 5 One may wonder at this stage why it is a *function* of the hearer to respond to tokens of sentences in given ways. In the case where the hearer is a waiter or waitress, it is clearly one of the hearer's functions to bring what the speaker requests. In the case of coming to believe the content of S, it is the function of the hearer because (1) *beliefs that p* certainly form a reproductively established family, and (2) believing the content of the sentence, in normal circumstances, is something selected for—the hearer believes the spoken content, rather than something else, because doing so has had a great variety of positive effects in past hearing-events.
- 6 I will refer to adaptationism as a methodology (rather than a theory) and wonder whether or not it is correct (rather than whether it is true) in order to avoid controversies over whether or not adaptationism constitutes a theory. I take it as uncontroversial that adaptationism is *at least* a methodology (a set of tools for dealing with a certain class of problems—see e.g. KITCHER 1985), even if some might dispute that it is a full-fledged theory (whether of a first- or higher-order). Resolving this issue is beyond the scope of the present paper. Thanks to an anonymous reviewer for this journal for raising this issue.
- 7 The alleles and fitnesses are as follows: A, S and C are initially present; most members of the population are AA; AS is fitter than AA; SS is lethal; C is recessive to A; CS is less fit than AA; CC is the most fit pair. In such a population, C is eliminated, and so the most fit pair CC is also eliminated, despite the fact that it was present initially. (see KITCHER 1985, p215)
- 8 The (now classic) example GOULD/VRBA discuss is the flippers of sea turtles. The flippers, they claim, are adaptations for swimming that also serve reasonably well as diggers, an activity for which they have been exapted. See MILLIKAN 1993, pp41–50, for a criticism of this particular case.
- 9 "Most of what the brain now does to enhance our survival lies in the domain of exaptation—and does not allow us to make hypotheses about the selective path of human history." (1982, p13)
- 10 SELZER 1993 is an entire book devoted to analyzing the *writing style* of this paper.

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